Higher-order linear PDE: Inverse operator method

$$D_x^n = \frac{\partial^n}{\partial x^n}$$

$$D_y^n = \frac{\partial^n}{\partial y^n}$$

$$\frac{\partial^2 z}{\partial x \partial y} = D_x D_y$$

$$F(D_x, D_y)\, z = f(x,y)$$

$$F(D_x, D_y)\, z = 0$$

$$\Rightarrow \quad (D_x - m_1 D_y)(D_x - m_2 D_y)\, z = 0$$

where m_1 and m_2 are roots of A.E.

Let $(D_x - m_1 D_y)\, z = 0$

$$\Rightarrow \frac{\partial z}{\partial x} - m_1 \frac{\partial z}{\partial y} = 0 \Rightarrow p - m q = 0$$

Lagrange's A.E is

$$\frac{dx}{1} = \frac{dy}{-m_1} = \frac{dz}{0}$$

First two fractions:

$$dy + m_1 dx = 0$$

Integrating, $y + m_1 x = c_1$

and $\frac{dz}{0} = \frac{dx}{1} \Rightarrow z = c_2$

$\therefore z = \phi(y + m_1 x)$ is a soln of

$(D_x - m_1 D_y)\, z = 0$.

* Let $z = \phi(y+mx)$ be a solution of

$F(D_x, D_y)z = 0$, then

$$D_x^r z = m\,\phi^{r}(y+mx)$$

$$D_y z = \phi^{r}(y+mx)$$

$$\therefore \quad F(D_x, D_y)z = m\,\phi'(y+mx) + \phi'(y+mx)$$

$$\Rightarrow (m+1)\phi'(y+mx) = 0$$

$$m+1=0 \qquad \underline{m=-1}$$

The A-E is simply obtained by replacing $D_x z$ by m and $D_y z$ by 1 in

$$F(D_x, D_y)z = 0.$$

<u>Ex</u>:

$$\frac{\partial^4 z}{\partial x^4} - \frac{\partial^4 z}{\partial y^4} = 0$$

A.E, $m^4 - 1 = 0 \Rightarrow m = \pm 1, \pm i$

G.S $\quad z = \phi_1(y+x) + \phi_2(y-x) + \phi_3(y+ix) + \phi_4(y-ix)$

<u>Ex 2</u>: Repeated root problem

$$(D_x^2 + 8D_x D_y + 16 D_y^2)z = 0$$

A.E $\quad m^2 + 8m + 16 = 0 \Rightarrow m = -4, -4$

G.S $\quad z = \phi_1(y-4x) + x\phi_2(y-4x).$

3 Solve $\left(D_x^2 - D_x D_y - 2D_y^2\right) z = 2x + 3y$

A.E, $m^2 - m - 2 = 0$

$\therefore m = -1, 2$

$z_c = \phi_1(y - x) + \phi_2(y + 2x)$.

$$\therefore z_p = \frac{1}{D_x^2 - D_x D_y - 2D_y^2}(2x + 3y)$$

$$= \frac{1}{D_x^2} \frac{1}{1 - \frac{D_y}{D_x} - \frac{2D_y^2}{D_x^2}}(2x + 3y)$$

$$= \frac{1}{D_x^2}\left(1 + \frac{D_y}{D_x} + \cdots\right)(2x + 3y)$$

$$= \frac{1}{D_x^2}\left\{2x + 3y + \frac{1}{D_x}(0 + 3)\right\} = \frac{1}{D_x^2}(2x + 3y + 3x)$$

$$= \frac{5}{6}x^3 + \frac{3x^2 y}{2}$$

$\therefore z = z_c + z_p$

Solve $\left(D_x^2 + 4D_x D_y + 4D_y^2\right) z = e^{2x + y}$

A.E, $m^2 + 4m + 4 = 0$ $m = -2, -2$

$z_c = \phi_1(y - 2x) + x\,\phi(y - 2x)$

$$z_p = \frac{1}{D_x^2 + 4D_x D_y + 4D_y^2} e^{2x + y} = \frac{1}{4 + 4.2 + 4.1} e^{2x}$$

$$= \frac{1}{16} e^{2x}$$

G.S. $z = z_c + z_p$.

Case III: $F(D_x, D_y) = \sin(ax+by)$ or $\cos(ax+by)$

$$D_x^2 \sin(ax+by) = -a^2 \sin(ax+by)$$
$$D_y^2 \sin(ax+by) = -b^2 \sin(ax+by)$$
$$D_x D_y \sin(ax+by) = -ab \sin(ax+by)$$

Ex: $(D_x^2 - 5D_xD_y + 4D_y^2)z = \sin(2x+y)$

A.E, $m^2 - 5m + 4 = 0 \Rightarrow m = 1, 4$

$$z_c = \phi(y+x) + \phi_2(y+4x)$$

$$z_p = \frac{1}{D_x^2 - 5D_xD_y + 4D_y^2} \sin(2x+y)$$
$$= \frac{1}{-2^2 + 5\times 2 + 4(-1)^2} \sin(2x+y)$$
$$= \frac{1}{2} \sin(2x+y)$$

Solve: $(D_x^3 - 4D_x^2 D_y + 4D_xD_y^2)z = 4[\cos(2x+y)]$

$m = 0, 2, 2$

$$z_c = \phi_1(y) + \phi_2(y+2x) + x\phi_3(y+2x)$$

$$z_p = \frac{1}{D_x^2 \cdot D_x - 4D_xD_y \cdot D_x + 4D_xD_y^2} 4\cos(2x+y)$$
$$= x\frac{1}{3D_x^2 - 8D_yD_x + 4D_y^2} 4\cos(2x+y)$$

$$z_p = x^2 \frac{1}{6D_x - 8D_y} 4\cos(2x+y) \quad (*)$$

$$= \frac{4x^2}{2} \frac{3D_x - 4D_y}{9D_x^2 - 16D_y^2} \cos(2x+y)$$

$$= 2x^2 \frac{-3\sin(2x+y)\cdot 2 + 4\sin(2x+y)}{+9(-2^2) - 16(-1^2)}$$

$$= 2x^2 \frac{-6\sin(2x+y) + 4\sin(2x+y)}{-36+16}$$

$$= 2x^2 \frac{-10\sin(2x+y)}{-20}$$

$$\boxed{z_p = x^2 \sin(2x+y)}$$

<u>Alternative:</u> Since $D_x \neq 0$, put $D_x = 2$, $D_y = 1$

in $(*)$

$$z_p = \frac{4x^2}{2} \cdot \frac{1}{3\times 2 - 4\cdot 1} \int \cos\theta \, d\theta \qquad \boxed{\theta = 2x+y}$$

$$= 2x^2 \frac{1}{+2} \sin\theta$$

$$\boxed{z_p = x^2 \sin(2x+y)}$$

Rule 4: $f(D_x, D_y) z = e^{ax+by} \sin(cx+dy)$

Ex: $(D_x^2 - D_x D_y - 6D_y^2) z = e^{2y} \sin 3x$

A.E, $m^2 - m - 6 = 0 \Rightarrow m = -3, 2$

$$z_c = \phi_1(y-3x) + \phi_2(y+2x)$$

$$z_p = \frac{1}{D_x^2 - D_x D_y - 6D_y^2} e^{2y} \sin(3x)$$

$$= e^{2y} \frac{1}{D_x^2 - D_x(D_y+2) - 6(D_y+2)^2} \sin 3x$$

$$= e^{2y} \frac{1}{D_x^2 - D_x D_y - 2D_x - 6D_y^2 - 24D_y - 24} \sin 3x$$

$$= e^{2y} \frac{1}{-9 - 0 - 2D_x - 0 - 24D_y - 24} \sin 3x$$

$$= e^{2y} \frac{1}{2D_x + 2D_y + 33} \sin 3x$$

$$= e^{2y} \frac{2D_x + 2D_y - 33}{(2D_x + 2D_y)^2 - 33^2} \sin 3x$$

$$= e^{2y} \frac{6\cos 3x - 33 \sin 3x}{4(-3^2) - 33^2} \sin 3x$$

$$= e^{2y} \frac{2\cos 3x - 11 \sin 3x}{375}$$

$$z = z_c + z_p$$

$$z_p = \text{R.P. of } \frac{e^{i(x+y)}}{4}\left(x^2 - \frac{3ix}{2} - \frac{13}{8}\right)$$

$$= \text{" "} \frac{\cos(x+y) + i\sin(x+y)}{4}\left\{\left(x^2 - \frac{13}{8}\right) - \frac{3ix}{2}\right\}$$

$$= \frac{1}{4}\left(x^2 - \frac{13}{8}\right)\cos(x+y) + \frac{3x}{2}\sin(x+2y)$$

$$\therefore z = z_c + z_p$$

H.T

1. $(D_x^2 + 2D_xD_y + D_y^2)z = x^2 + xy + y^2$
2. $(D_x^2 + 5D_xD_y + 6D_y^2)z = 5e^{x-y}$
3. $(D_x - 2D_y)^2(D_x + 3D_y)z = e^{2x+y}$
4. $(D_x^3 - 7D_xD_y^2 - 6D_y^3)z = \sin(x+2y)$
5. $(D_x^3 - 4D_x^2D_y + 4D_xD_y^2)z = \sin(2x+y)$
6. $(D_x^3 + D_x^2D_y - D_xD_y^2)z = e^x\cos 2y$.
7. $(D_x^2 + 3D_xD_y + 2D_y^2)z = x+y$
8. $(D_x^2 - 6D_xD_y + 9D_y^2)z = 12x^2 + 36xy$
9. $(D_x^2 - D_xD_y - 6D_y^2) = x^2\sin(3x+y)$
10. $(D_x^2 - 5D_xD_y + 6D_y^2) = e^{x-y}xy$.